Prof. Pierre Duysinx
Quentin Sneessens
A&M Automotive Engineering
University of Liège

October 2026

MECA0527 EV, HEV, and FC vehicles

DESIGN AND SIMULATION OF BATTERY ELECTRIC VEHICLE AND ASSESSMENT OF RANGE EXTENDER ADDITION

In 2013, German automaker BMW launched the revolutionary BMW i3, marking one of the first mass-produced premium Battery Electric Vehicles (BEVs) on the market.



Figure 1 - BMW i3 (2013)

This homework consists of simulating the performance and energy consumption of this vehicle, including its energy consumption on a standardized driving cycle.

At first it is required to complete the set of all necessary information to elaborate an effective numerical model. The vehicle is then simulated to estimate and assess its performance and energy consumption.

A major limitation of early BEVs was their driving range, practically restricted to around 160 km for the first-generation i3. To overcome range anxiety, BMW offered a Range Extender (REX) version. This architecture integrates a downsized twin-cylinder 28 kW internal combustion engine (ICE), coupled exclusively to a generator. The engine

converts fuel into electricity on the go to maintain the battery charge, thereby extending the driving range.

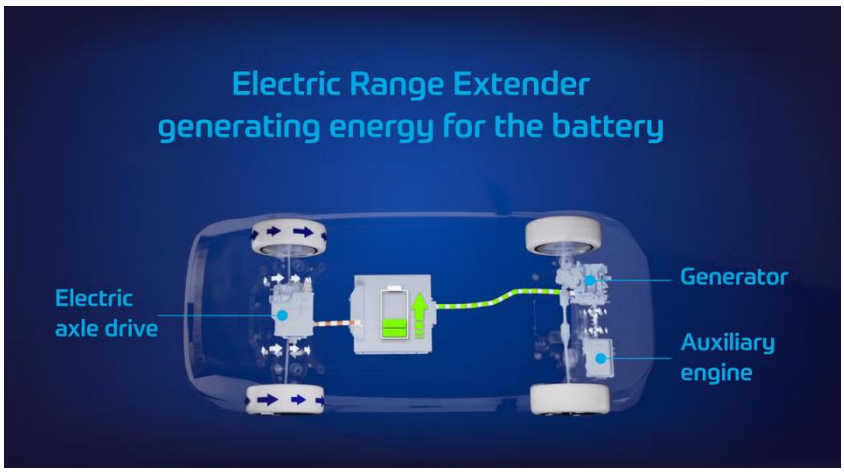


Figure 2 - Range Extender Schematic

In the second part of the homework, students will model and simulate this range extender, establish an energy management strategy, and discuss its implementation. They will assess the range gain on a standardized driving cycle and conclude on the viability and relevance of this technology for this class of vehicles. Once the battery was depleted, the REX version was reported to be unable to sustain highway speeds, which raised safety concerns.

Finally, an optional bonus part investigates a fuel cell range extender as an alternative to the current REX.

EXPECTED OUTCOMES

HW PART 1: BEV Version

1. Complete the set of data describing the vehicle and its components (some are given in the Appendix). Justify all your assumptions and the resulting numerical values.
2. Road resistance: plot your results on clear figures (force and power diagrams). Consider different slopes of 0%, 5% and 10%.
3. Model and plot the traction curves (tractive force and power curves) of the electric motor.
4. Net force and acceleration capability: plot the net force available for acceleration and the resulting acceleration capability as a function of speed, for the three slopes.
5. Simulate the performance of the BEV vehicle. Investigate and discuss the following questions:
    a. Top speed
    b. Maximum slope from rest and at 80 kph
    c. Acceleration times. Consider the following cases:
        i. Elasticity time from 50 to 90 kph
        ii. Acceleration time from 0 to 100 kph

iii. Acceleration time to cover 0 to 400 m

6. Simulate the energy consumption of the vehicle for BEV version:
    a. Compute the energy consumption [kWh/km] at constant speed at 30, 50, 90 and 120 kph.
    b. Draw the energy consumption [kWh/km] curves as a function of the driving speed from 5 kph to 150 kph. Consider constant speeds.
    c. Estimate the energy consumption when performing the WLTP driving cycle (see Appendix E). Deduce the vehicle range if the trip is based on this WLTP cycle.

HW PART 2: REX Version

7. Model the range extender: engine torque curve and BSFC map, generator efficiency, and the resulting electrical power delivered to the battery.
8. Implement an energy management strategy for the REX version (how is the thermal engine used to produce electricity). The main goal is to maximize range.
9. Simulate the performance of the REX version, accounting for its additional mass and modified aerodynamics, in two situations: (1) battery charged, and (2) battery depleted with the REX as the only energy source. Investigate and discuss:
    a. Top speed
    b. Maximum slope from rest and at 80 kph
    c. Acceleration times. Consider the following cases:
        i. Elasticity time from 50 to 90 kph
        ii. Acceleration time from 0 to 100 kph
        iii. Acceleration time to cover 0 to 400 m
10. Simulate the energy consumption of the vehicle for both versions:
    a. Draw the energy consumption [kWh/km and L/100km] curves as a function of the driving speed from 5 kph to 150 kph. Consider constant speeds.
    b. Estimate the energy consumption when performing the WLTP driving cycle (see Appendix E). Deduce the vehicle range if the trip is based on this WLTP cycle.
11. Once the battery was depleted, the REX version was reported to be unable to sustain highway speeds, particularly uphill (eg: V_{max} limited to 70kph) [12]. Using your model, explain this behaviour.

HW PART 3: Fuel Cell REX Version (Bonus)

12. Replace the ICE range extender with a fuel cell system. Size the fuel cell power and the hydrogen storage, and compare the sustainable highway speed, the mass and the WLTP range with the ICE REX.

PRACTICAL INFORMATION

The work is carried out in groups of two students. Each group submits two reports of 30 pages maximum each, in PDF format:

- Report 1 (Part 1): due **Thursday, November 12, 2026**, at 23:59 (Brussels time)
- Report 2 (Part 2, and optional Part 3): due **Wednesday, December 23, 2026**, at 23:59 (Brussels time)

The reports must include a clear description of the data collection, the assumptions made, the equations used, the complete results (possibly in appendix), a critical analysis of the results, and the developed computer code (possibly in appendix).

Reports are submitted by email to p.duysinx@uliege.be and q.sneessens@uliege.be.

In compliance with the rules of the Faculty of Engineering of ULiège, any use of AI tools in developing parts of the code or sections of the report must be clearly identified by the authors.

Appendix A: Vehicle characteristics

GEOMETRY, MASS AND INERTIA PROPERTIES

- Length / Width / Height (mm): 3999 x 1775 x 1597
- Wheelbase: 2570 mm
- Curb weight of the standard version (without a driver): 1195 kg
- Driver mass: 75 kg
- Weight distribution between front and rear axles: 47/53 for standard version & 44/56 for REX
- Elevation of the centre of gravity: to be estimated
- Electric motor inertia: to be estimated from technical data

AERODYNAMICS

- $C_X =$ 0.29 for standard version & 0.30 for REX
- $S =$ 2.38 m²
- $\rho_{air} =$ 1.23 kg/m³

TYRES AND TRANSMISSION

- Standard tire size: 155/70 R 19
- Effective rolling radius: to be estimated
- Transmission efficiency: to be estimated
- Rolling resistance coefficients: $f_{\mathrm{rr}} = 0.008 + 0.4 \times 10^{-7}\, V^2$ [-] (with V in km/h)
- Transmission to the rear wheels
- Reduction ratio: $i =$ 9.665

MOTOR AND BATTERY SPECIFICATIONS

- Motor power: 125 kW peak / 75 kW nominal @ 4800 RPM
- Motor torque: 250 Nm
- Regeneration: up to 50 kW
- Battery: 22.60 kWh gross

RANGE EXTENDER

- Engine power: 28 kW @ 5000 RPM
- Engine torque: 56 Nm @ 4500 RPM
- Device weight: 120 kg
- Number of cylinders of the engine: 2
- Engine size: 647 cm³
- Stroke / bore of the engine: 66 / 79 mm
- Generator power: 26.6 kW
- Fuel tank capacity: 9 L
- Reduction ratio between the ICE and the generator: $i_{gen} =$ 2.0

Appendix B: Electric motor

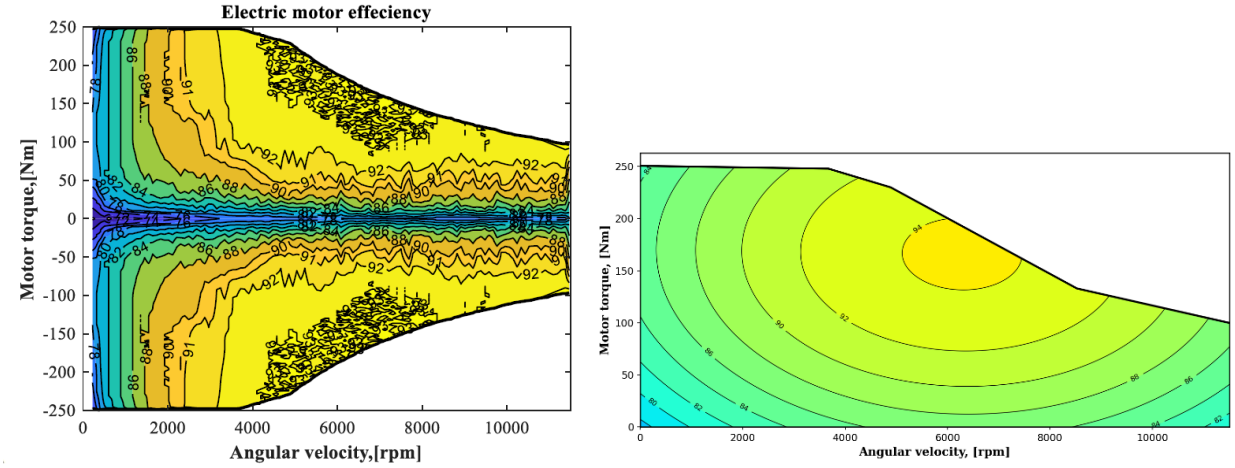


Figure 3 - Electric Motor Efficiency Map, real and approximated

This efficiency map can be approximated using Golverk method. To get a decent approximation, it was decided to use a polynomial of order 2, which reads:

$$\begin{aligned}\eta(N,T) = {} & C_0 \\ & + C_1 \cdot N + C_2 \cdot T \\ & + C_3 \cdot N^2 + C_4 \cdot N \cdot T + C_5 \cdot T^2\end{aligned}$$

The coefficients have been computed and are given below:

Coefficient	Value	Units
C_0	+8.079E-1	[-]
C_1	+1.827E-5	[1/rpm]
C_2	+6.760E-4	[1/Nm]
C_3	-1.087E-9	[1/rpm^2]
C_4	+2.434E-9	[1/Nm/rpm]
C_5	-1.864E-6	[1/Nm^2]

Table 1 - Empirical coefficients to generate the efficiency map

Appendix C: REX Engine BSFC map

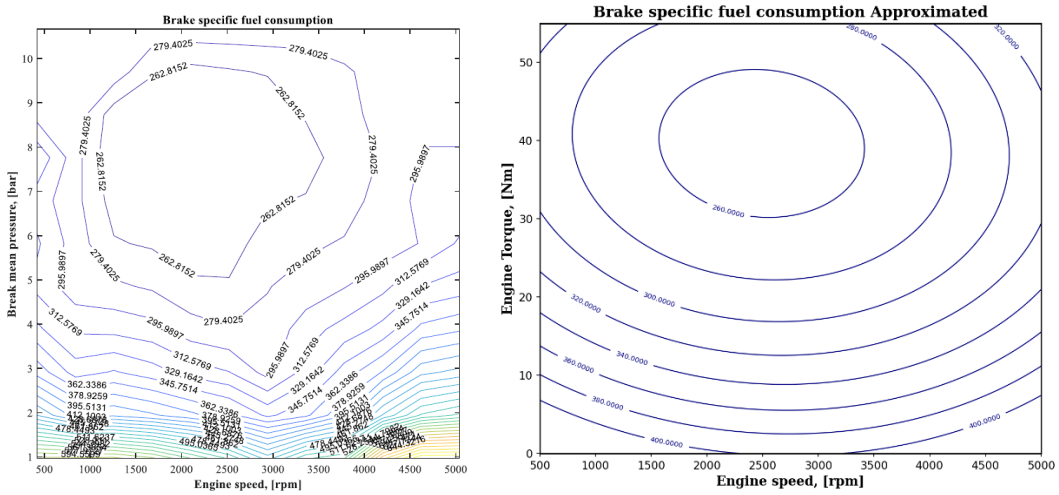


Figure 4 - REX Engine BSFC Map, real and approximated

This bsfc map can be approximated using Golverk method. To get a decent approximation, it was decided to use a polynomial of order 2, which reads:

$$bsfc(N,T) = C_0 + C_1 \cdot N + C_2 \cdot T + C_3 \cdot N^2 + C_4 \cdot N \cdot T + C_5 \cdot T^2$$

The coefficients have been computed and are given below:

Coefficient	Value	Units
C_0	+4.726E+2	[g/kWh]
C_1	-5.426E-2	[g/kWh/rpm]
C_2	-7.736E+0	[g/kWh/Nm]
C_3	+9.845E-6	[g/kWh/rpm^2]
C_4	+1.308E-4	[g/kWh/Nm/rpm]
C_5	+9.346E-2	[g/kWh/Nm^2]

Table 2 - Empirical coefficients to generate the BSFC map

Appendix D: REX Generator

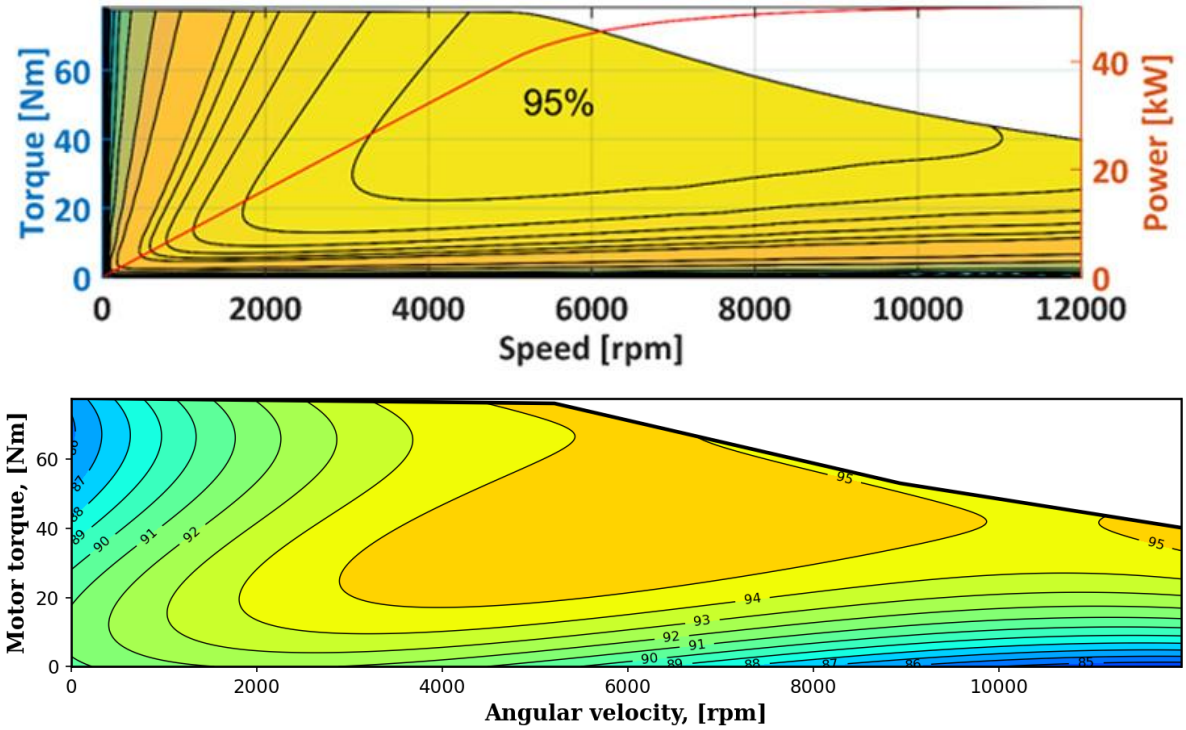


Figure 5 - REX Generator Efficiency map, real and approximated

This efficiency map can be approximated using Golverk method. To get a decent approximation, it was decided to use a polynomial of order 3, which reads:

$$\begin{aligned}\eta(N,T) = {} & C_0 \\ & + C_1 \cdot N + C_2 \cdot T \\ & + C_3 \cdot N^2 + C_4 \cdot N \cdot T + C_5 \cdot T^2 \\ & + C_6 \cdot N^3 + C_7 \cdot N^2 \cdot T + C_8 \cdot N \cdot T^2 + C_9 \cdot T^3\end{aligned}$$

The coefficients have been computed and are given below:

Coefficient	Value	Units
C_0	+9.071E-1	[-]
C_1	+1.358E-5	$[1/\mathrm{rpm}]$
C_2	+1.149E-3	$[1/\mathrm{Nm}]$
C_3	-3.662E-9	$[1/\mathrm{rpm}^2]$
C_4	+6.036E-7	[1/Nm/rpm]
C_5	-6.612E-5	$[1/\mathrm{Nm}^2]$
C_6	+1.734E-13	[1/rpm³]
C_7	-1.434E-11	$[1/\mathrm{rpm}^2/\mathrm{Nm}]$
C_8	-3.817E-9	[1/Nm²/rpm]
C_9	+5.683E-7	$[1/\mathrm{Nm}^3]$

Table 3 - Empirical coefficients to generate the efficiency map

Appendix E: WLTP Driving cycle

Please refer to the Excel file on *Teams* platform for the precise speed – time values.

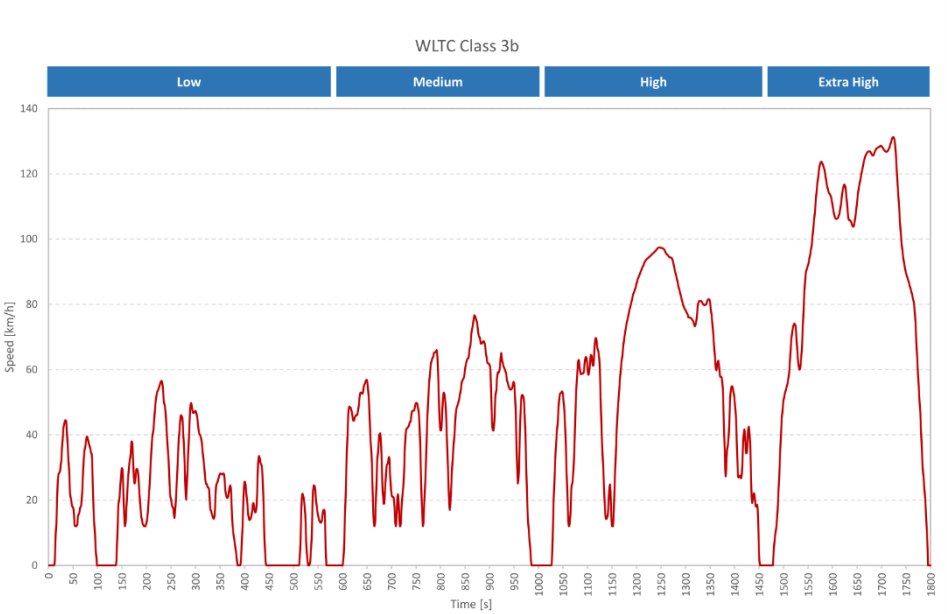


Figure 6 - WLTP Cycle

References

1. 2022 BMW i3 Hatchback Electric Vehicle full range specs. s. d. Consulté 1 septembre 2026. https://www.automobile-catalog.com/make/bmw/i3/i3_hatch/2022.html.
2. 2022 BMW i3 tire sizes (up to July 2022 for Europe). s. d. Consulté 1 septembre 2026. https://www.automobile-catalog.com/tire/2022/2966765/bmw_i3.html.
3. Bonfitto, Angelo, Sanjarbek Ruzimov, et Malika Keldiyarova. s. d. « Development of Control Strategy for BMW I3 Range Extender ».
4. Holjevac, Nikola, Federico Cheli, et Massimiliano Gobbi. 2019. « A Simulation-Based Concept Design Approach for Combustion Engine and Battery Electric Vehicles ». *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering* 233(7):1950-67. doi:10.1177/0954407018777350.
5. Horsepower/Torque Curve for 2022 BMW i3 (aut. 1) (model up to July 2022 for Europe). Detailed engine characteristics. s. d. Consulté 1 septembre 2026. https://www.automobile-catalog.com/curve/2022/2966765/bmw_i3.html.
6. Jeong, Jongryeol, Wonbin Lee, Namdoo Kim, Kevin Stutenberg, et Aymeric Rousseau. 2017. « Control Analysis and Model Validation for BMW I3 Range Extender ». in *SAE Technical Paper*.
7. Miri, Ilyès, Abbas Fotouhi, et Nathan Ewin. 2021. « Electric Vehicle Energy Consumption Modelling and Estimation—A Case Study ». *International Journal of Energy Research* 45(1):501-20. doi:10.1002/er.5700.
8. Noga, Marcin, Paweł Gorczyca, et Radosław Hebda. 2020. « The Effects of Use of the Range Extender in a Small Commercial Electric Vehicle ». *Automotive Experiences* 4(1):5-19. doi:10.31603/ae.4137.
9. Pielecha, Ireneusz. 2020. « Control Algorithms for a Range Extender Vehicle with an Combustion Engine ». *Combustion Engines* 183(4):3-10. doi:10.19206/CE-2020-401.
10. Sedik, Raouache Youcef, Layadi Toufik Madani, Becherif Mohamed, et Zaoui Fares. 2023. « Modeling and Dynamic Simulation of an Electric Vehicle ». P. 1-6 in *2023 1st International Conference on Renewable Solutions for Ecosystems: Towards a Sustainable Energy Transition (ICRSEtoSET)*. Djelfa, Algeria: IEEE.
11. Xu, Chuyuan. 2023. « Research on Range Extended Electric Vehicle Based on Cruise ». *Journal of Physics: Conference Series* 2479(1):012012. doi:10.1088/1742-6596/2479/1/012012.
12. Torregrossa, M. (2014, December 18). BMW i3 – Le « range extender » en manque de puissance ? Automobile Propre. https://www.automobile-propre.com/breves/bmw-i3-range-extender-manque-de-puissance/